

Unit 3

Assignment Questions

1.	Consider the following page-reference string: 2 3 2 1 5 2 4 5 3 2 5 2. How many page fault occur in FIFO, LRU and Optimal page replacement algorithms, assuming three frames?	Written
2.	Consider the logical address space of 256 pages with 4 KB of Size , mapped onto the Physical memory of 64 frames. How many bits are required for logical address and physical address?	Written
3.	Consider a computer system with a 32 bit logical address and 4 KB page Size. The system supports upto 512 MB of Physical Memory. How many number of pages are there?	Written
4.	Create a kernel module which takes the target_pid as command line argument and calculate the total number of pages in logical space and number of pages allocated in resident/primary memory. Execute and Put the screenshots of the make , module insert, kernel messages Hint: Headers <pre>#include <linux/mm.h> #include <linux/pid.h></pre> Inbuilt structures to use <pre>struct task_struct *task; struct mm_struct *mm = task ->mm</pre> Variables to access <pre>mm->total_vm #for total pages get_mm_rss (mm) #for resident memory pages</pre>	Practical